

Modeling & Verification 2025

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- Labelled transition systems with time
- Timed CCS; syntax and semantics
- Timed Automata; syntax and semantics

Turing Award

$$a.0 \mid (b.0 + c.0)$$



Robin Milner

1991, CCS

$$\langle a \rangle [b] ff$$



Amir Pnueli

1996, Temporal Logic

$$a.0 \mid (b.0 + c.0) \models \langle a \rangle [b] ff$$



Alan Emerson



Ed Clarke



Tony Hoare

1980, CSP



Joseph Sifakis

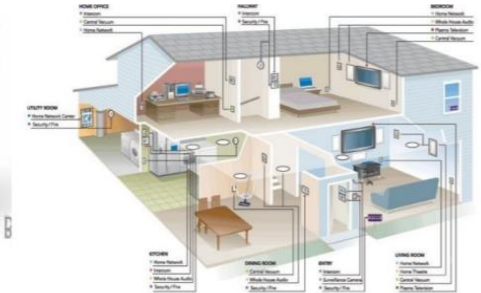
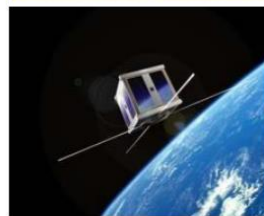
2007, Model Checking

Need for introducing Time Features

- Timeout in Alternating Bit protocol:
 - In CCS timeouts were modelled using nondeterminism.
 - Enough to prove that the protocol is safe.
 - Maybe too abstract for certain questions (What is the average time to deliver the message?).
- Many real-life systems depend on timing:
 - Real-time controllers (production lines, computers in cars, railway crossings).
 - Embedded systems (mobile phones, remote controllers, digital watch).
 - ...

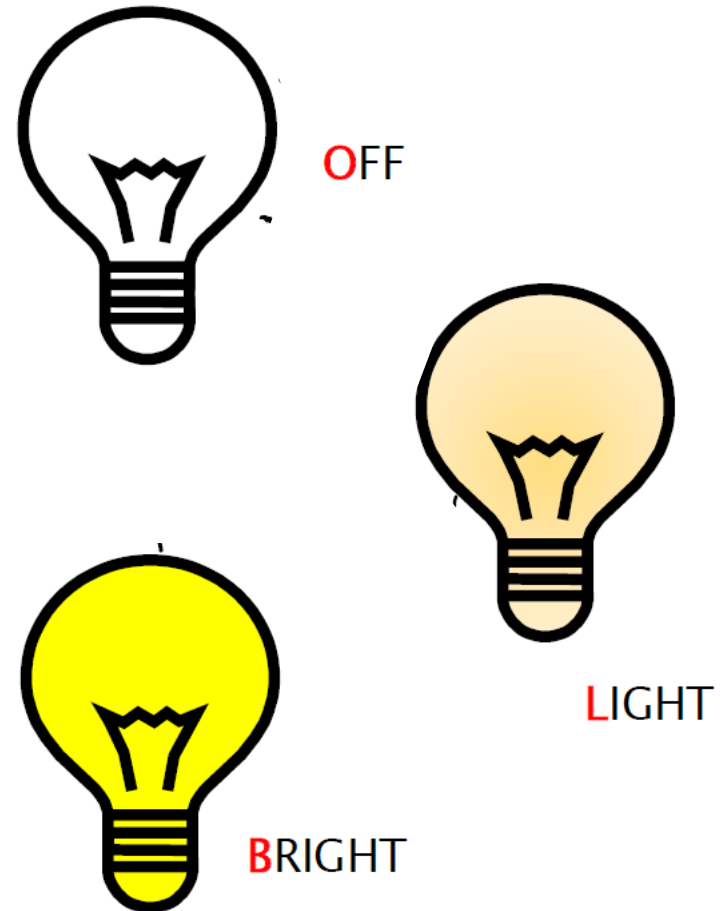
Embedded Systems

- 80% of all software is embedded
- Demands for **increased functionality** with **minimal resources**
- Requires multitude of skills
 - Software construction
 - hardware platforms,
 - communication
 - testing & verification
- **Goal:**
Give a qualitative lift to
current industrial practice
!!!!!!



A Light Switch

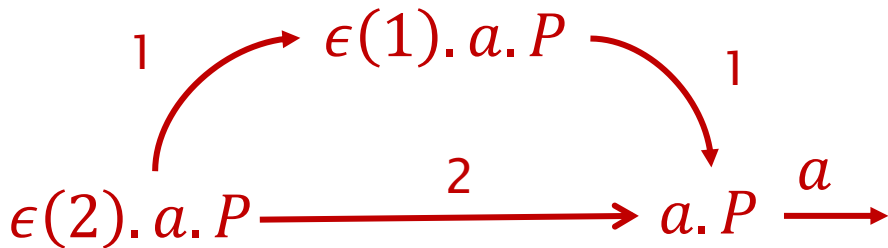
- If off and **pr**essed once light will turn on
- If **pr**essed again *soon after* then the light will become brighter
- Otherwise light will turn off
- Also, if **pr**essed when light is bright the light will turn off



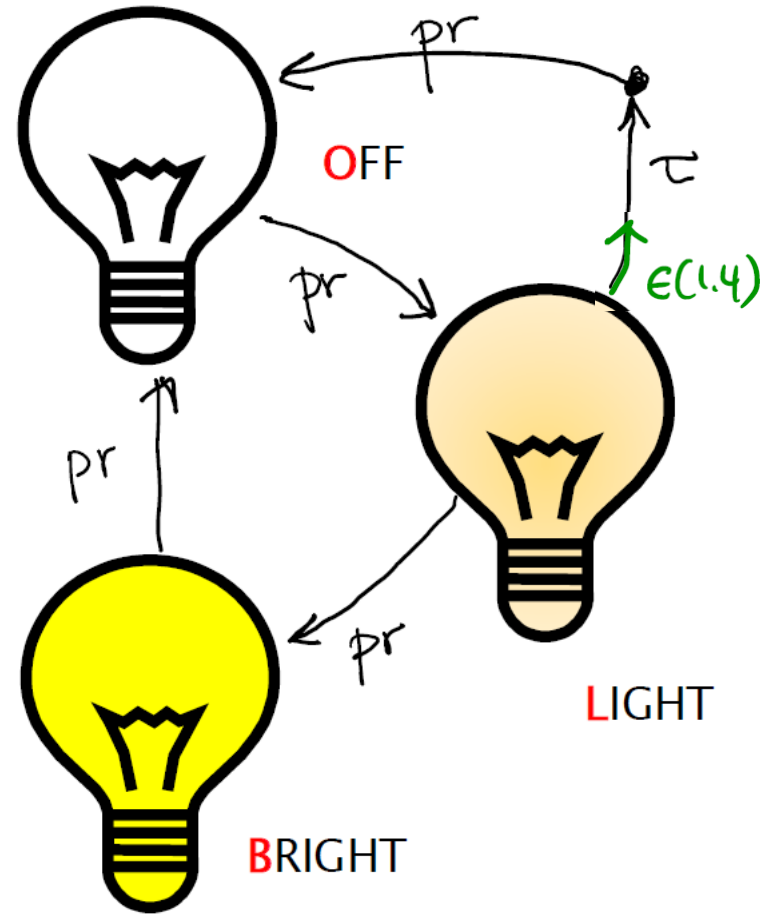
A Light Switch

$$\begin{aligned} O &\stackrel{\text{def}}{=} \text{pr}.L \\ B &\stackrel{\text{def}}{=} \text{pr}.O \\ L &\stackrel{\text{def}}{=} \end{aligned}$$

$$\tau.\text{pr}.O + \text{pr}.B$$



$$\epsilon(d).P \quad d \in \mathbb{R}_{\geq 0}$$



A Light Switch

$$L \xrightarrow{1.1} \text{pr} \rightarrow B$$

$$L \xrightarrow{2/\cancel{2}} \text{pr} \rightarrow \dots$$

$$L \xrightarrow{0.8} \text{pr} \rightarrow B$$

$$L \xrightarrow{0.8} \epsilon(0.6). \tau. \text{pr}. O + \text{pr}. B$$

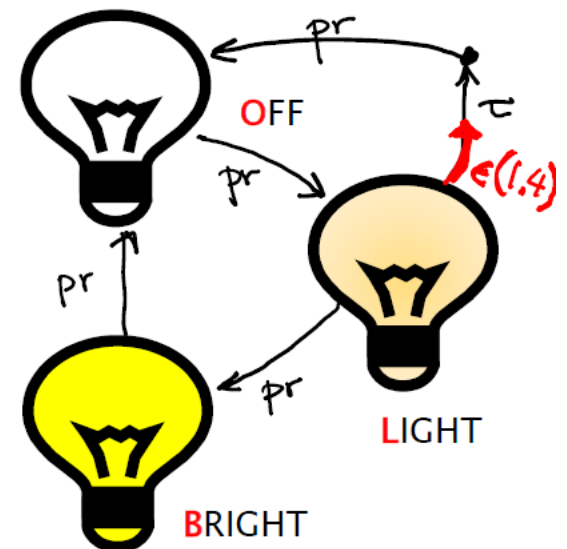
Off = pr. Light

Bright = pr. Off

Light = $\epsilon(1.4). \tau. \text{pr}. \text{Off} + \text{pr}. \text{Bright}$

$$L \xrightarrow{1.4} \text{pr} \rightarrow B$$

$$L \xrightarrow{1.4} \tau \rightarrow \text{pr}. O$$



Timed Transition System

Timed (labelled) transition system (TLTS)

TLTS is a triple $(Proc, Act, \{\xrightarrow{a} \mid a \in Act\})$ where

- $Proc$ is a set of states (or processes),
- $Act = N \cup \mathbb{R}^{\geq 0}$ is a set of **actions** (consisting of **labels** and **time-elapsing steps**), and
- for every $a \in Act$, $\xrightarrow{a} \subseteq Proc \times Proc$ is a binary relation on states called the transition relation.

We write

- $s \xrightarrow{a} s'$ if $a \in N$ and $(s, s') \in \xrightarrow{a}$, and
- $s \xrightarrow{d} s'$ if $d \in \mathbb{R}^{\geq 0}$ and $(s, s') \in \xrightarrow{d}$.

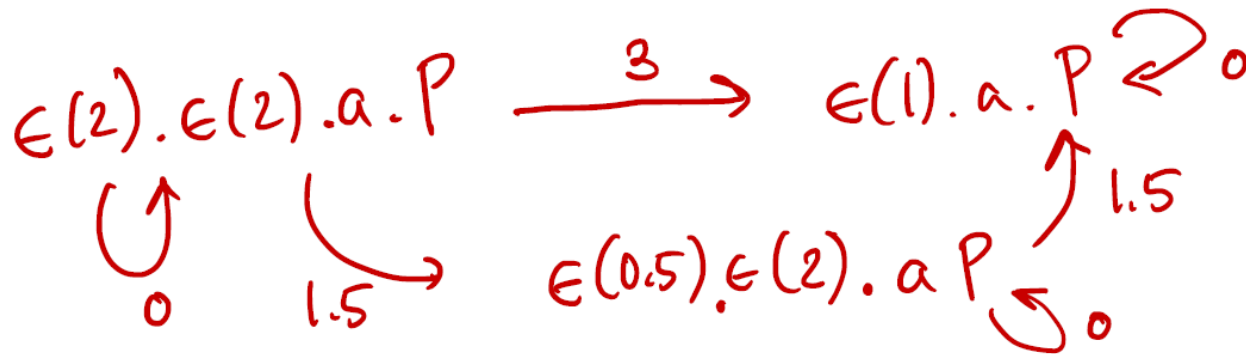
Requirements to TLTS

Sanity Requirements

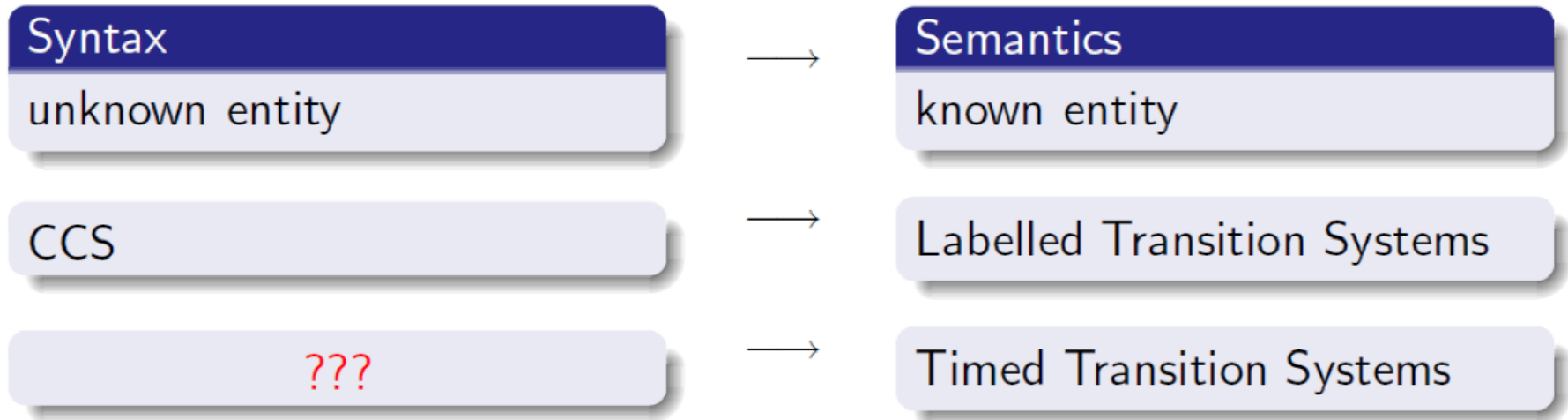
Time additivity: If $s \xrightarrow{d} s'$ and $0 \leq d' \leq d$ then $s \xrightarrow{d'} s'' \xrightarrow{d-d'} s'$ for some state s'' ;

Zero delay: $s \xrightarrow{0} s$ for all states s ;

Time determinism: If $s \xrightarrow{d} s'$ and $s \xrightarrow{d} s''$ then $s' = s''$.



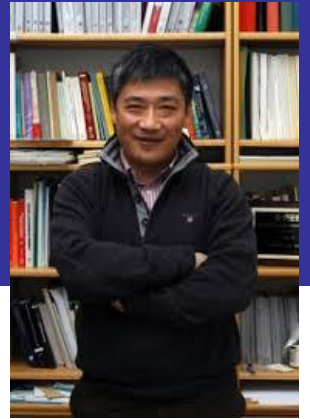
How to describe TLTS



TCCS [Yi'90]:
CCS extended with delays.

Timed Automata [Alur, Dill'90]:
Finite-state automata equipped with clocks.

Timed CCS



Wang Yi
Uppsala

CCS Syntax

Process expressions:

$P :=$	K		process constants ($K \in \mathcal{K}$)
	$\alpha.P$		prefixing ($\alpha \in Act$)
	$\sum_{i \in I} P_i$		summation (I is an arbitrary index set)
	$P_1 P_2$		parallel composition
	$P \setminus L$		restriction ($L \subseteq \mathcal{A}$)
	$P[f]$		relabelling ($f : Act \rightarrow Act$) such that
			• $f(\tau) = \tau$ • $f(\bar{a}) = \overline{f(a)}$

$$P_1 + P_2 = \sum_{i \in \{1,2\}} P_i$$

$$Nil = 0 = \sum_{i \in \emptyset} P_i$$

CCS program

A collection of **defining equations** of the form $K \stackrel{\text{def}}{=} P$ where $K \in \mathcal{K}$ is a process constant and P is a process expression.

$$\text{ACT} \quad \frac{}{\alpha.P \xrightarrow{\alpha} P} \qquad \text{SUM}_j \quad \frac{P_j \xrightarrow{\alpha} P'_j}{\sum_{i \in I} P_i \xrightarrow{\alpha} P'_j} \quad j \in I$$

$$\text{COM1} \quad \frac{P \xrightarrow{\alpha} P'}{P|Q \xrightarrow{\alpha} P'|Q} \qquad \text{COM2} \quad \frac{Q \xrightarrow{\alpha} Q'}{P|Q \xrightarrow{\alpha} P|Q'}$$

$$\text{COM3} \quad \frac{P \xrightarrow{a} P' \quad Q \xrightarrow{\bar{a}} Q'}{P|Q \xrightarrow{\tau} P'|Q'}$$

$$\text{RES} \quad \frac{P \xrightarrow{\alpha} P'}{P \setminus L \xrightarrow{\alpha} P' \setminus L} \quad \alpha, \bar{\alpha} \notin L \qquad \text{REL} \quad \frac{P \xrightarrow{\alpha} P'}{P[f] \xrightarrow{f(\alpha)} P'[f]}$$

$$\text{CON} \quad \frac{P \xrightarrow{\alpha} P' \quad K \xrightarrow{\alpha} P'}{K \stackrel{\text{def}}{=} P}$$

TCCS=CCS + Delay Prefix

Let $d \in \mathbb{R}^{\geq 0}$ and let P be a process then $\epsilon(d).P$ is the process which after a delay of d time units will behave like P .

Rules for Delay Prefix

We expect the following transitions

- $\epsilon(d).P \xrightarrow{d} P$
- $\epsilon(d).P \xrightarrow{d'} \epsilon(d - d').P$ for $d' \leq d$
- $\epsilon(d).P \xrightarrow{d+d'} P'$ if $P \xrightarrow{d'} P'$.

SOS Rules for TCCS (delays)

DEL₁

$$\frac{P \xrightarrow{d'} P'}{\epsilon(d).P \xrightarrow{d+d'} P'}$$

DEL₂

$$\frac{}{\epsilon(d).P \xrightarrow{d'} \epsilon(d-d').P} \quad d' \leq d$$

CON_d

$$\frac{P \xrightarrow{d} P'}{K \xrightarrow{d} P'} \quad K =_{\text{def}} P$$

ACT_d

$$\frac{}{\alpha.P \xrightarrow{d} \alpha.P} \quad \alpha \neq \tau$$

SUM_d

$$\frac{P \xrightarrow{d} P' \quad Q \xrightarrow{d} Q'}{P + Q \xrightarrow{d} P' + Q'}$$

REL_d

$$\frac{P \xrightarrow{d} P'}{P[f] \xrightarrow{d} P'[f]}$$

RES_d

$$\frac{P \xrightarrow{d} P'}{P \setminus L \xrightarrow{d} P' \setminus L}$$

A Light Switch

DEL₁

$$\frac{P \xrightarrow{d'} P'}{\epsilon(d).P \xrightarrow{d+d'} P'}$$

DEL₂

$$\frac{}{\epsilon(d).P \xrightarrow{d'} \epsilon(d-d').P} \quad d' \leq d$$

CON_d

$$\frac{P \xrightarrow{d} P'}{K \xrightarrow{d} P'} \quad K =^{\text{def}} P$$

ACT_d

$$\frac{}{\alpha.P \xrightarrow{d} \alpha.P} \quad \alpha \neq \tau$$

SUM_d

$$\frac{P \xrightarrow{d} P' \quad Q \xrightarrow{d} Q'}{P + Q \xrightarrow{d} P' + Q'}$$

REL_d

$$\frac{P \xrightarrow{d} P'}{P[f] \xrightarrow{d} P'[f]}$$

RES_d

$$\frac{P \xrightarrow{d} P'}{P \setminus L \xrightarrow{d} P' \setminus L}$$

L $\xrightarrow{0.8}$

Off = pr. Light
 Bright = pr. Off
 Light = $\epsilon(1.4). \tau. \text{pr. Off} + \text{pr. Bright}$

A Light Switch

NO RULE CAN BE APPLIED DUE to SIDE-CONDITION IN ACT₂.

$$\begin{array}{c}
 \text{DEL}_1 \quad \frac{P \xrightarrow{d'} P'}{\epsilon(d).P \xrightarrow{d+d'} P'} \quad \text{DEL}_2 \quad \frac{}{\epsilon(d).P \xrightarrow{d'} \epsilon(d-d').P} \quad d' \leq d \\
 \text{CON}_d \quad \frac{P \xrightarrow{d} P'}{K \xrightarrow{d} P'} \quad K =^{\text{def}} P \quad \text{ACT}_d \quad \frac{}{\alpha.P \xrightarrow{d} \alpha.P} \quad \alpha \neq \tau \quad \text{SUM}_d \quad \frac{P \xrightarrow{d} P' \quad Q \xrightarrow{d} Q'}{P + Q \xrightarrow{d} P' + Q'} \\
 \text{REL}_d \quad \frac{P \xrightarrow{d} P'}{P[f] \xrightarrow{d} P'[f]} \quad \text{RES}_d \quad \frac{P \xrightarrow{d} P'}{P \setminus L \xrightarrow{d} P' \setminus L}
 \end{array}$$

$$\begin{array}{l}
 \text{Del}_1 \quad \frac{\tau.\text{pr.}0 \xrightarrow{0.b} ?}{\epsilon(1.4).\tau.\text{pr.}0 \xrightarrow{2} ??} \quad \frac{}{\text{pr.}B \xrightarrow{2} \text{pr.}B} \text{ACT}_2 \\
 \frac{\epsilon(1.4).\tau.\text{pr.}0 \xrightarrow{2} ??}{\epsilon(1.4).\tau.\text{pr.}0 + \text{pr.}B \xrightarrow{2} ??} \text{SUM}_d \\
 \frac{\epsilon(1.4).\tau.\text{pr.}0 + \text{pr.}B \xrightarrow{2} ??}{L \xrightarrow{2} ??} \text{CON}_d
 \end{array}$$

Off = pr. Light
 Bright = pr. Off
 Light = $\epsilon(1.4).\tau.\text{pr.Off} + \text{pr.Bright}$

A Light Switch | A User

$(\text{Light} \mid \text{User}) \setminus pr$

$\xrightarrow{0.3}$

$(\epsilon(1.4). \tau.pr.\text{Off} + pr.\text{Bright} \mid \overline{pr}.\text{User}) \setminus pr$

$\xrightarrow{\tau}$

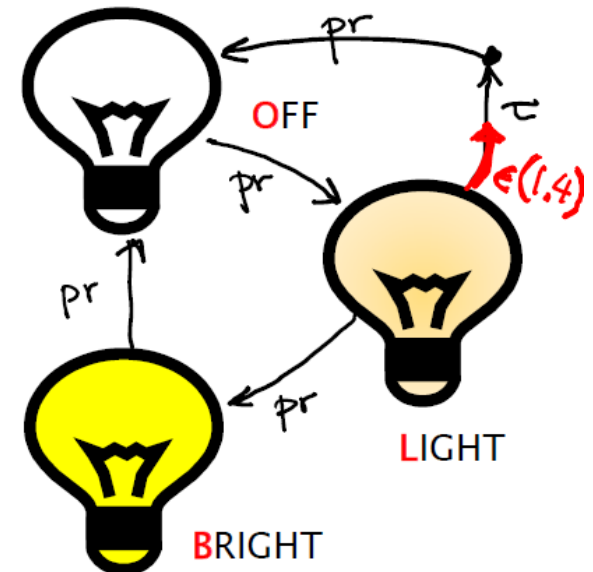
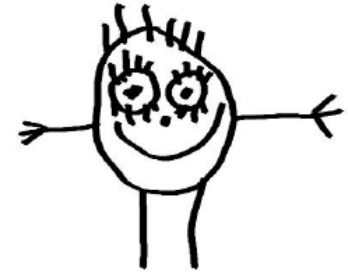
$(\text{Bright} \mid \text{User}) \setminus pr$

$\text{User} = \epsilon(0.3). \overline{pr}.\text{User}$

$\text{Off} = pr.\text{Light}$

$\text{Bright} = pr.\text{Off}$

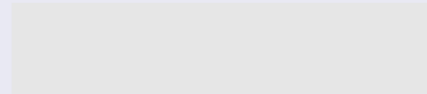
$\text{Light} = \epsilon(1.4). \tau.pr.\text{Off} + pr.\text{Bright}$



Parallel Composition

Delay for Parallel Composition

$$\frac{P \xrightarrow{d} P' \quad Q \xrightarrow{d} Q'}{P \mid Q \xrightarrow{d} P' \mid Q'}$$



Parallel Composition

Maximal Progress:

If a process can evolve on its own, then it will do so without any further delay, i.e. if $P \xrightarrow{\tau}$ then $P \not\xrightarrow{d}$ for any $d > 0$.

Delay for Parallel Composition

$$\frac{P \xrightarrow{d} P' \quad Q \xrightarrow{d} Q'}{P \mid Q \xrightarrow{d} P' \mid Q'}$$

..

Parallel Composition

Maximal Progress:

If a process can evolve on its own, then it will do so without any further delay, i.e. if $P \xrightarrow{\tau}$ then $P \not\xrightarrow{d}$ for any $d > 0$.

Delay for Parallel Composition

$$\frac{P \xrightarrow{d} P' \quad Q \xrightarrow{d} Q'}{P \mid Q \xrightarrow{d} P' \mid Q'} \quad \text{NoSync}(P, Q, d)$$

where $\text{NoSync}(P, Q, d)$ holds if for any $d' < d$ whenever $P \xrightarrow{d'} P'$ and $Q \xrightarrow{d'} Q'$ then $P' \mid Q' \not\xrightarrow{\tau}$

Light Switch | User

User	=	$\epsilon(0.3).\overline{pr}.User$
Off	=	$pr.Light$
Bright	=	$pr.Off$
Light	=	$\epsilon(1.4).\tau.pr.Off + pr.Bright$

Exercise: Complete the proof
below

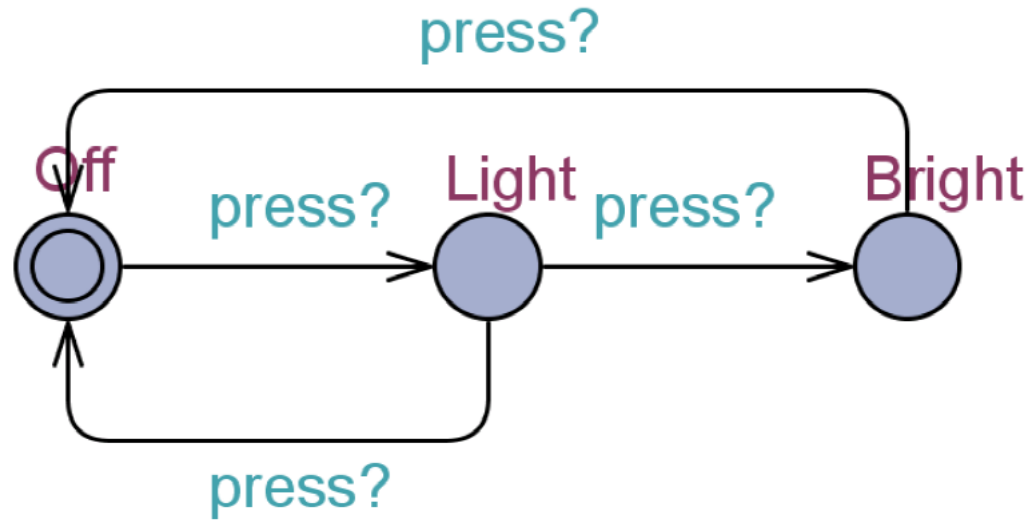


$$(Light \mid User) \setminus pr \xrightarrow{0.3} ??$$

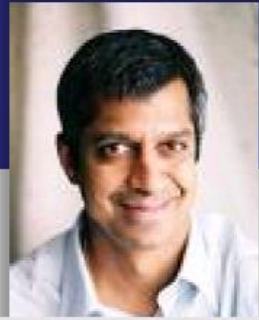
Timed Automata



A Dumb Light Controller



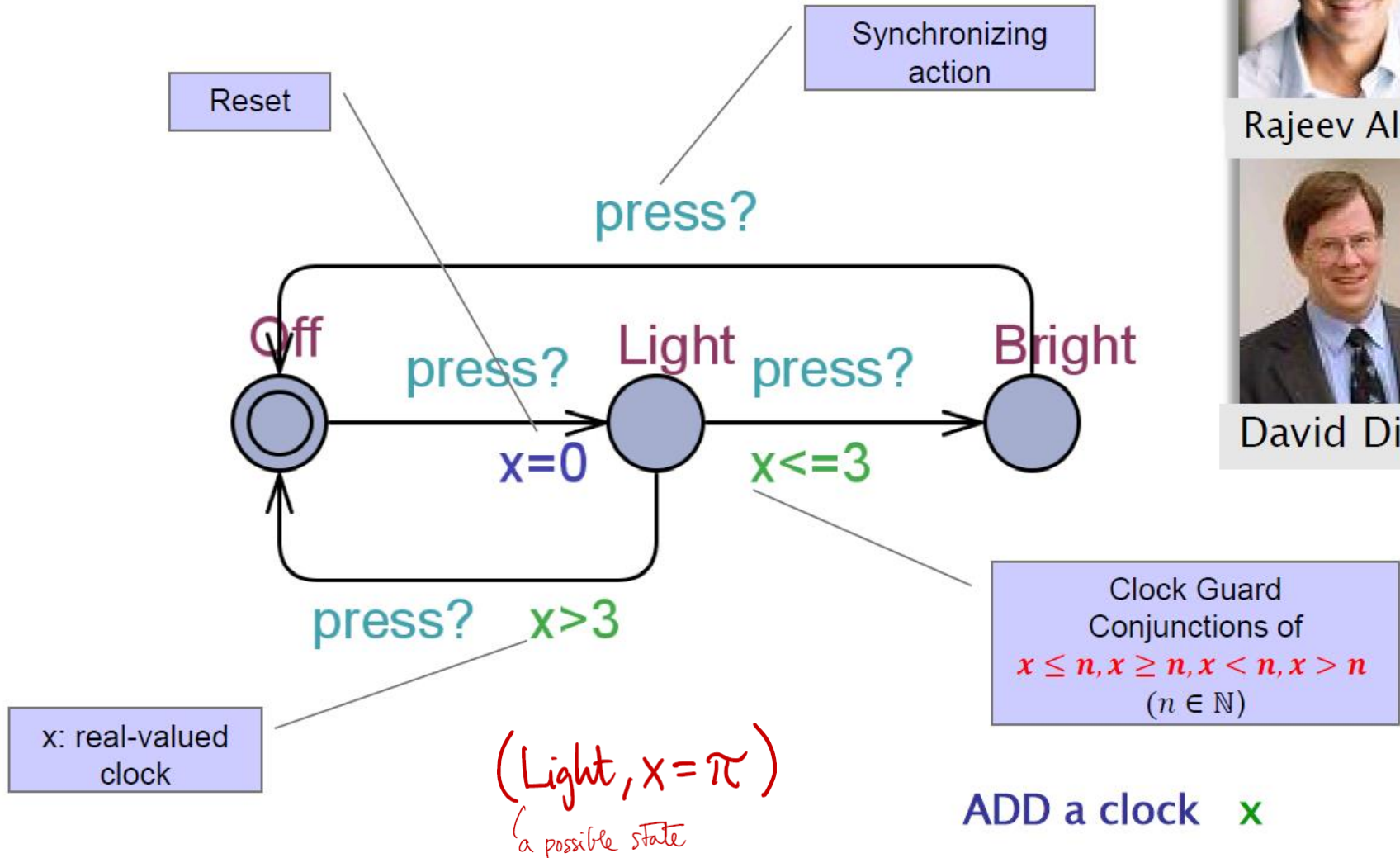
A Dumb Light Controller



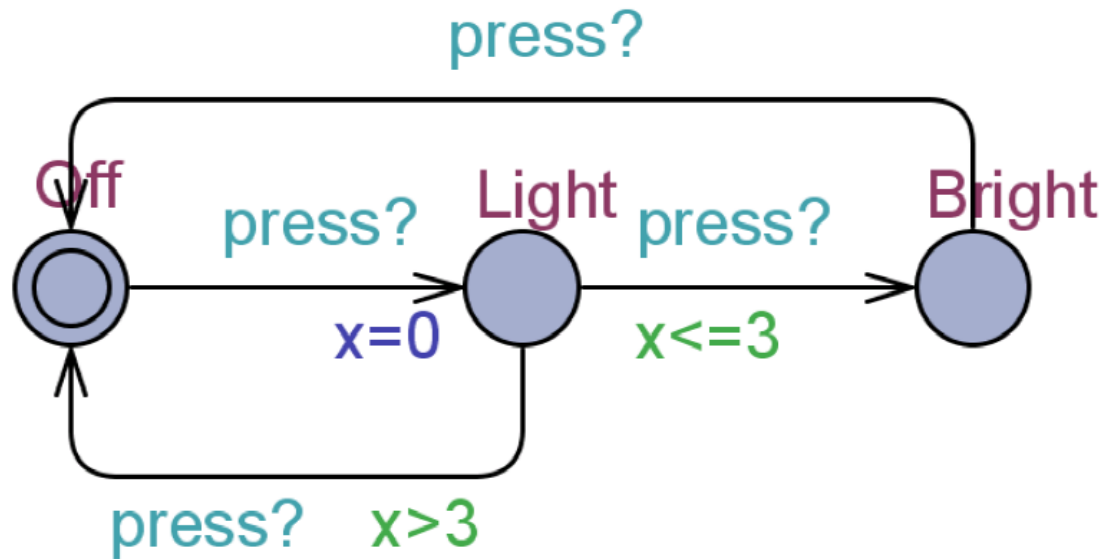
Rajeev Alur



David Dill



Timed Automata (Semantics)



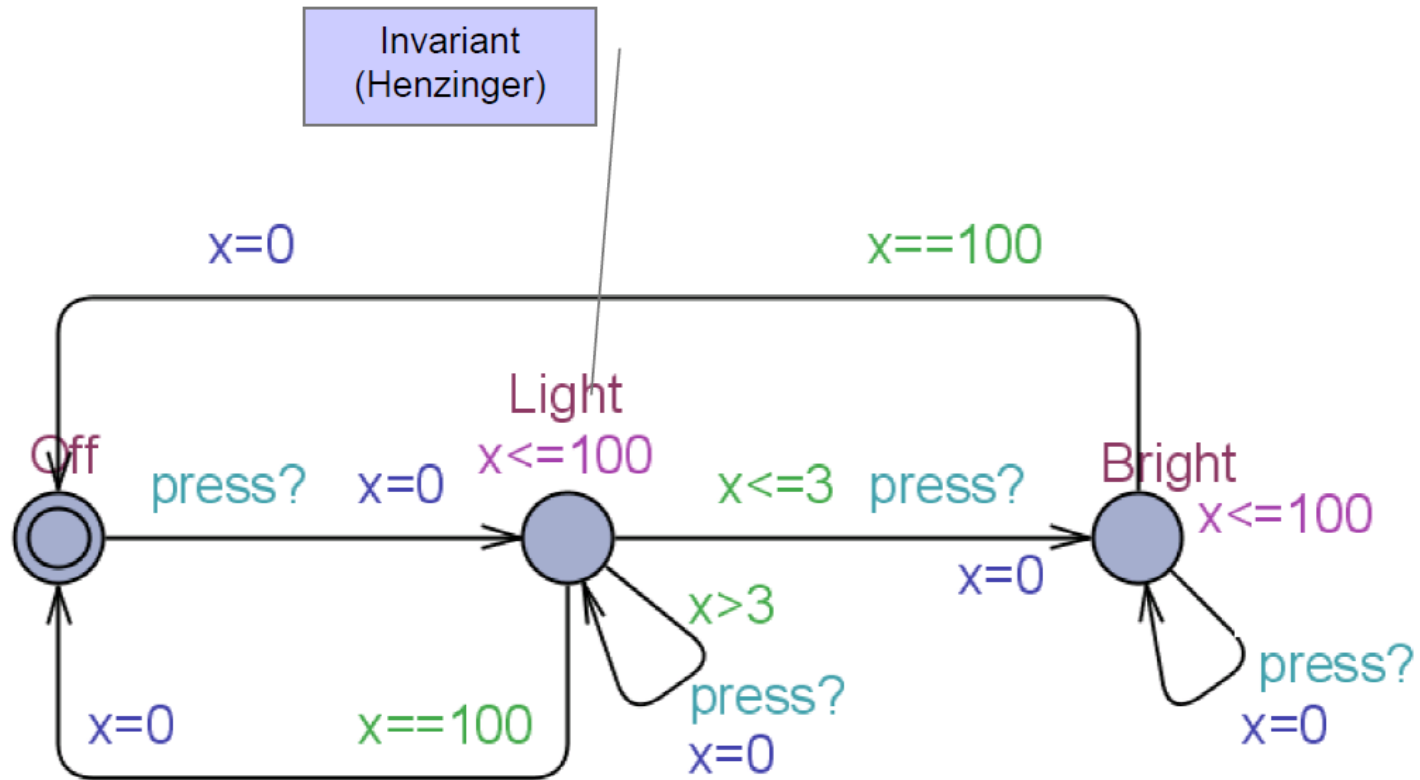
States:

$(loc, x = v)$ where $v \in \mathbb{R}$

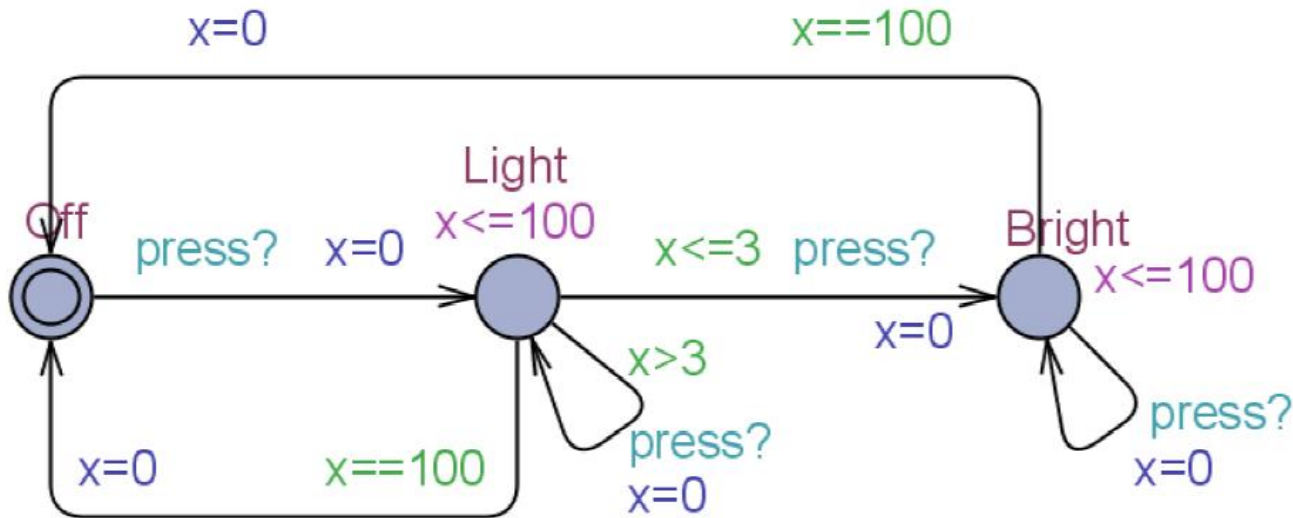
Transitions:

$(O, x=0) \xrightarrow{2.1} (O, x=2.1)$
 $\xrightarrow{pr?} (L, x=0)$
 $\xrightarrow{3.1} (L, x=3.1)$
 $\xrightarrow{pr?} (O, x=3.1)$

Intelligent Light Controller



Intelligent Light Controller



Transitions:

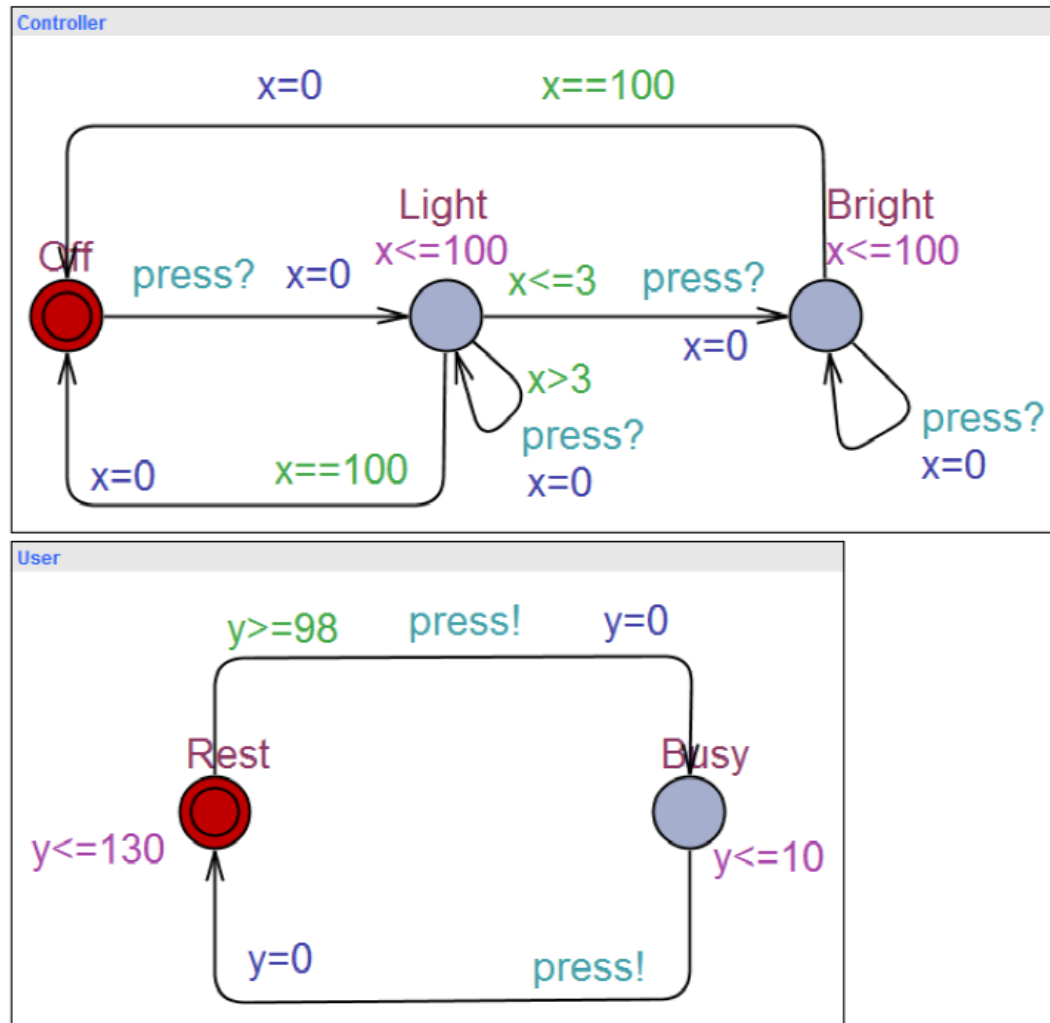
$(0, x=0) \xrightarrow{\text{press?}} (L, x=0) \xrightarrow{5.7} (L, x=5.7)$

$\xrightarrow{\text{press?}} (L, x=0)$

~~$\xrightarrow{102} ???$~~

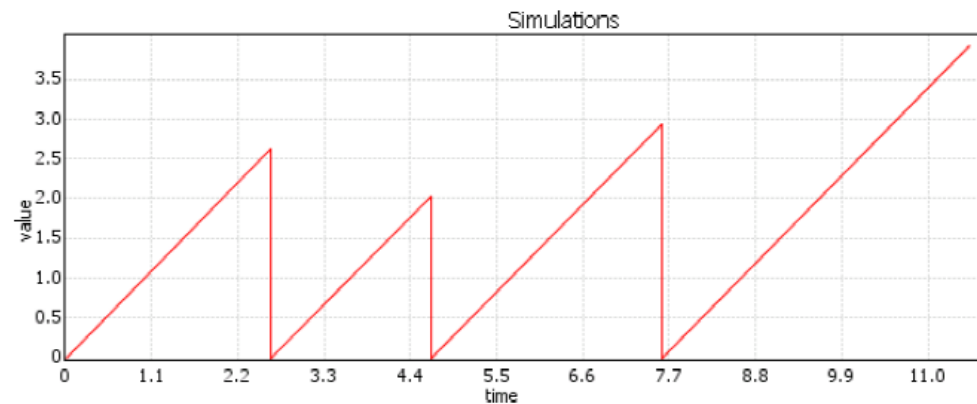
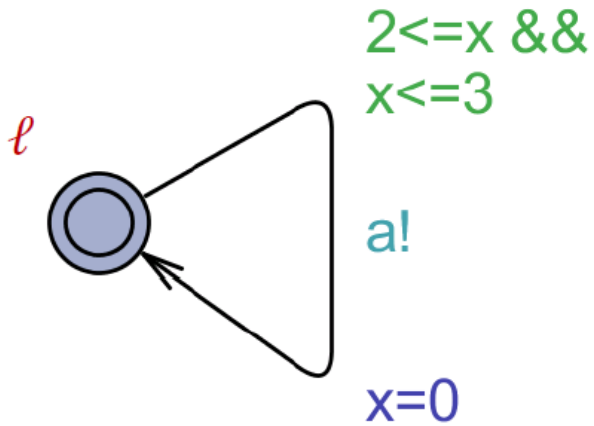
Not possible because of Inv in L

UPPAAL Demo

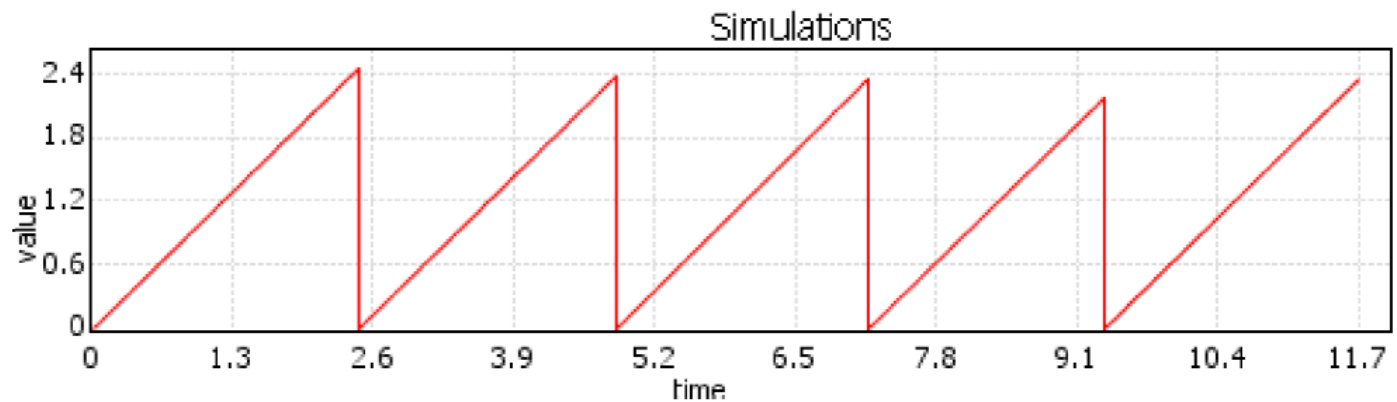
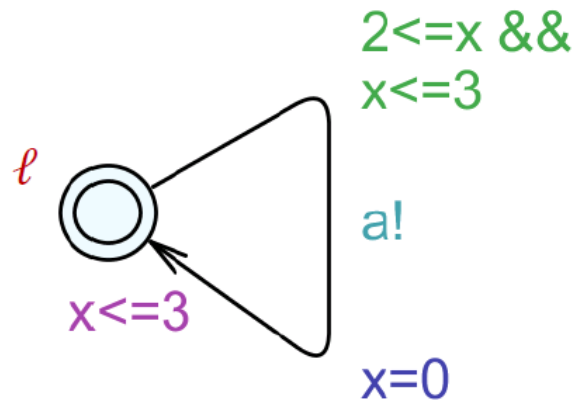


dumblight

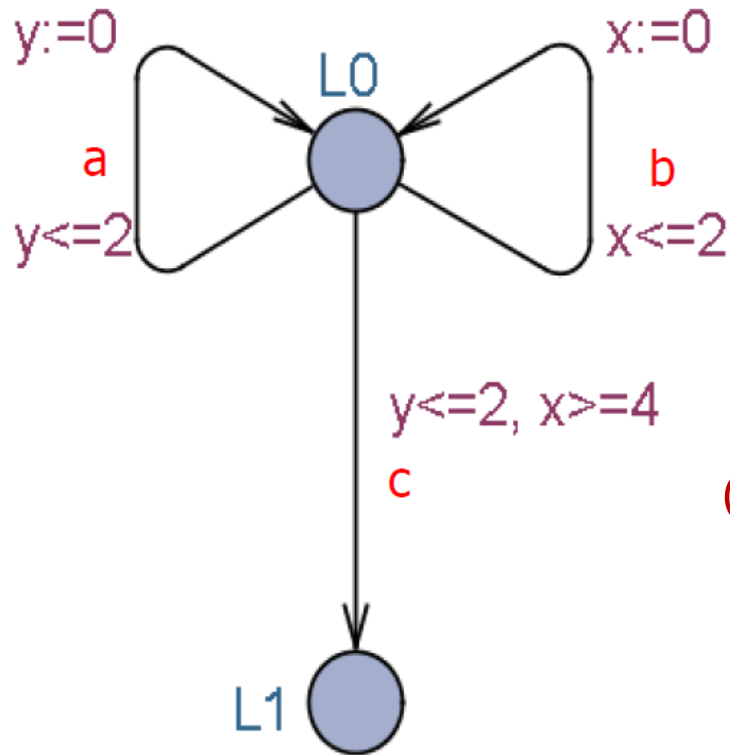
Example



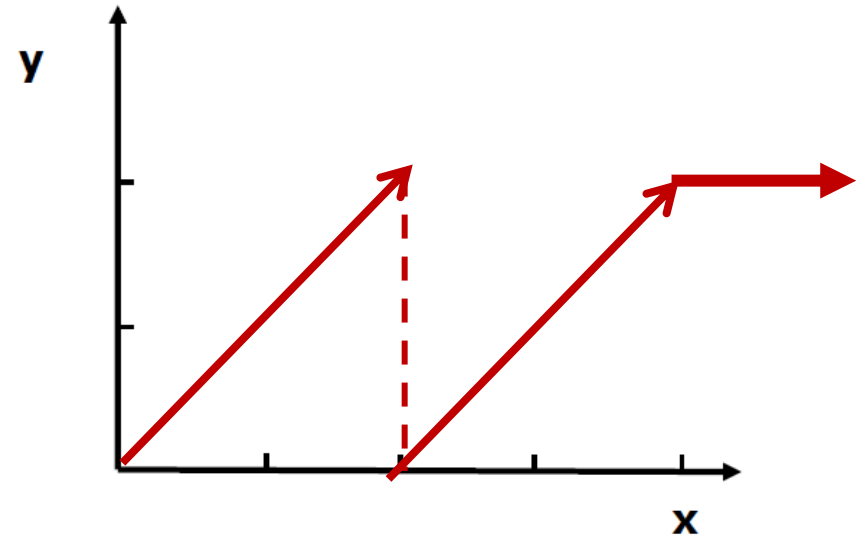
Example



Example



Is **L1** reachable ?



$(L_0, x = 0, y = 0) \xrightarrow{-\epsilon(2)}$
 $(L_0, x = 2, y = 2)$
 $\xrightarrow{-a}$
 $(L_0, x = 2, y = 0)$
 $\xrightarrow{-\epsilon(2)}$
 $(L_0, x = 4, y = 2)$
 $\xrightarrow{-c}$

Thank You !

www.uppaal.org
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